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Dept: BS-CS

Section: BCS-3D

Subject: COAL-LAB

Submitted To:

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Lab 4:

Question 1:

CODE	EXPLANATION
<pre>[org 0x0100] mov word [1200h], 0x1234 mov word [1202h], 0x5678 mov bx, 1200h mov ax, [bx] mov cx, [bx + 2] mov ax, 0x4c00 int 0x21</pre>	➤ In this program I store the value '1234h' at DS: 1200 and '5678' at DS:1202h. ➤ At line 4, BX register points to DS: 1200h. ➤ At line 5, I moved the value of DS:1200h into AX register via bx register. ➤ At line 6, I moved the value of DS:1202h into CX register via BX register.

Question 2:

Code	Explanation
<pre>[org 0x0100] xor ax, ax xor bx, bx mov bx, num1 add al, [bx] add al, [bx + 1] add al, [bx + 2] mov [num1 + 3], al mov ax, 0x4c00 int 0x21 num1: db 10, 20, 30, 0</pre>	➤ At line 2 and 3, we clean up AX and BX register. ➤ At line 4, BX points to starting address of data. ➤ At line 5, I add the starting value of num1 using indirect addressing (via BX register). ➤ At line 6, I add the second byte (value) using offsets ➤ At line 7, I added the third byte (value) using offset ➤ At line 8, I stored the final value into the last memory location.

Question 3:

Code	Explanation
<pre>[org 0x0100] xor ax, ax xor bx, bx mov al, [data] mov bl, [data + 1] add al, bl mov bl, [data + 2] add al, bl mov [data + 3], al mov ax, 0x4c00 int 0x21 data: db 7, 12, 20, 0</pre>	➤ At line 2 and 3, we clean up AX and BX register. ➤ At line 4, I load the first value of “data” (7) into AL register ➤ At line 5, I load the second value of “data” (12) into BL ➤ At line 6, I added AL and BL. ➤ At line 7, I load the third value of “data” (20) into BL register. ➤ At line 8, I added AL and BL register. ➤ At line 9, I stored the final value into 4th location.

Question 4:

Code	Explanation
<pre>[org 0x0100] xor ax, ax mov bx, data add ax, [bx] add bx, 2 add ax, [bx] add bx, 2 add ax, [bx] mov [result], ax mov ax, 0x4c00 int 0x21 data: dw 7, 12, 20 result: dw 0</pre>	➤ At line 2, I clean up the AX register (value updated to 0). ➤ At line 3, Bx is pointing to start location of data. ➤ At line 4, I added the first value of “data” using BX pointer and stored the value in ax register. ➤ At line 5, I moved to the next location. ➤ At line 6, I added the value of ax and bx and stored the value into ax register. ➤ At line 7, I again moved to the next location of data (20) using bx pointer. ➤ At line 8, I added the value of ax and bx and stored the value into ax register. ➤ At line 9, I stored the result into separate variable.

Question 5:

Code	Explanation
<pre>[org 0x0100] xor ax, ax mov cx, 5 mov si, data sum: add ax, [si] add si, 2 dec cx jnz sum mov [result], ax mov ax, 0x4c00 int 0x21 data: dw 7, 12, 20, 35, 9 result: dw 0</pre>	➤ At line 2, I cleanup the ax register (value updated to 0). ➤ Stored 5 value into cx register at line 3. ➤ 'Si register pointing to start of data at line 4. ➤ Line 5, start of loop. ➤ Line 6, add current value of si into ax register. ➤ Line 7, mov si to next value. ➤ Line 8, decrement counter by 1. ➤ Line 9, check condition "jump if not zero" until CX = 0. ➤ Line 10, store total sum into result variable.